Momentum Conservation – One of Nature's Most Strictly Enforced Laws

Synopsis: Along with energy and matter conservation, which you are perhaps familiar with from chemistry, the conservation of momentum law is one of Nature's laws that one has never seen or will ever see being violated. As a matter of fact, you can build a theory on the premise of this conservation law and make predictions based on the fact that you can count on this law to be observed.

Just as one can never create or destroy matter or energy (the total is always the same), nor can one create or destroy momentum – the total momentum before and after an event must be the same, i.e., conserved.

In this Note, I will present one of the most important equations in physics, which describes the conservation of momentum. You will observe a few examples of the conservation law in action and see how the equation summarizes the conservation phenomena.

When I say an event in the context of momentum conservation, I mean a collision – two bodies in motion coming into contact. When there is contact, that is a collision event. One of two things will happen: they will either collide and separate, or they will stick together and move as a unit. When they bounce, it's called an elastic collision. When they stick, it's called an inelastic collision.

We will begin with demonstrations of elastic collisions using the smart carts and Newton's Cradle.

Performance Expectations: Describe how the conservation of momentum equation describes collision events.

For PHY 151 — with Newton's laws: Derive momentum conservation for two colliding bodies from Newton's second and third laws, and show why it holds whatever the collision does to the kinetic energy.

Instructions and Questions

1) Carts that bounce.
   i) Watch the demonstration of colliding carts that don't stick after a collision. How is the conservation of momentum law manifested? Draw a diagram of before and after, showing the velocity vectors.

Before	After

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2) Newton's Cradle.
   i) Stare at the hypnotic bouncing back and forth of a row of hanging steel balls in Newton's Cradle. How is the conservation of momentum law manifested? Draw a diagram showing the velocity vectors before and after.

Before	After

3) The equation.
   i) Write the equations for momentum before and after.
   ii) Write the general momentum conservation equation when two bodies collide.

Before	After

4) The equation with numbers.
   Two smart carts on the level track, east positive. Cart A is 2.0 kg and cart B is 1.0 kg.

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i) Cart A moves east at 3.0 m/s and hits cart B, at rest. They stick. Write the total momentum before, and find the speed they move off with together.
ii) Same start, but this time they bounce and cart A is measured afterwards at 1.0 m/s east. How fast is cart B moving, and which way?
iii) Cart A at 3.0 m/s east meets cart B at 3.0 m/s west, and they stick. What happens?

Why the total cannot change.

During the contact the carts push on each other with equal and opposite forces (third law), $F_{AB} = -F_{BA}$, and they are in contact for the same time Δt. Newton's second law in the form $F = \Delta p/\Delta t$ gives $\Delta p_A = F_{AB}\,\Delta t$ and $\Delta p_B = F_{BA}\,\Delta t$, so $\Delta p_A = -\Delta p_B$ and the sum $p_A + p_B$ changes by nothing at all. Nothing in the argument mentions kinetic energy, which is why the total momentum is the same whether the carts bounce or stick — the kinetic energy is not.

i) Write the argument as three lines of algebra ending in $p_A + p_B = p_A' + p_B'$.
ii) For the sticking collision in section 4, calculate the kinetic energy before and after. Where did the difference go?

5) Looking ahead.

Just like energy conservation, momentum conservation follows the "zero sum" idea: all the changes in momenta add up to zero. Next we look at what a single body's change of momentum costs — a force, acting for a time.

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